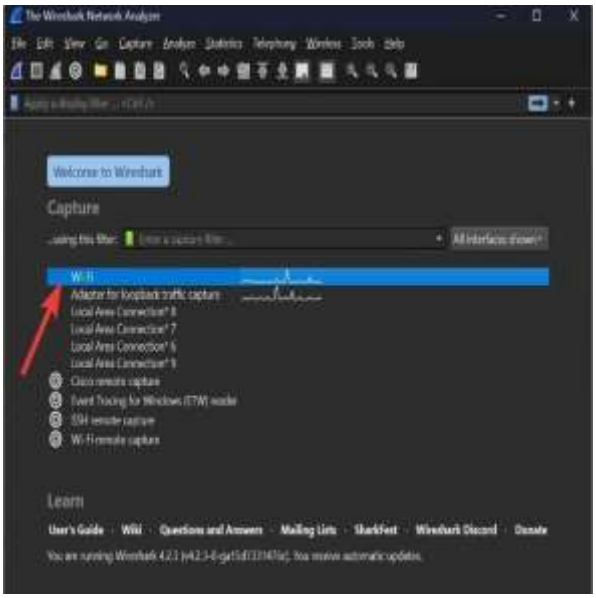
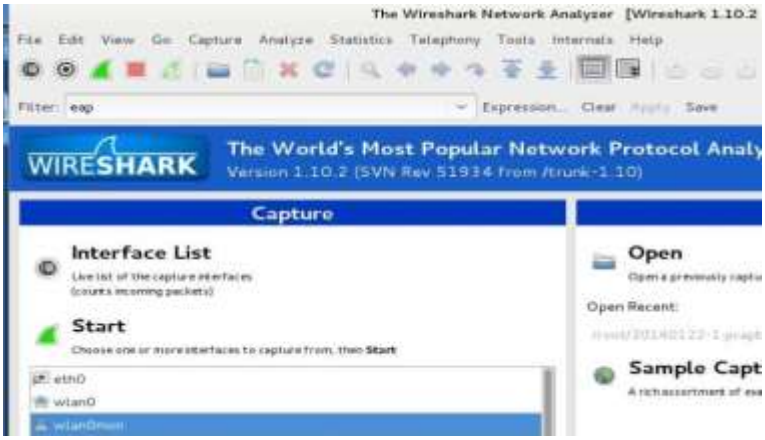
DAY-1 (NETWORKING)

TASK-1 Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.





TASK 1: Wireshark Packet Capture.

STEP:

1. Installed and opened Wireshark.
2. Selected the active network interface.
3. Started packet capture.
4. Visited https://www.instagram.com in the web browser.
5. Stopped the capture after generating network traffic.
6. Observed the Protocol column in the packet list.

DAY-1 (NETWORKING)

TASK -2 Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents

Packet 1: DNS

- OSI Layer: Layer 7 — Application
- Function: Resolves domain names into IP addresses.

Packet 2: TCP

- OSI Layer: Layer 4 — Transport
- Function: Provides reliable and ordered data delivery.

Packet 3: IPv4/IPv6

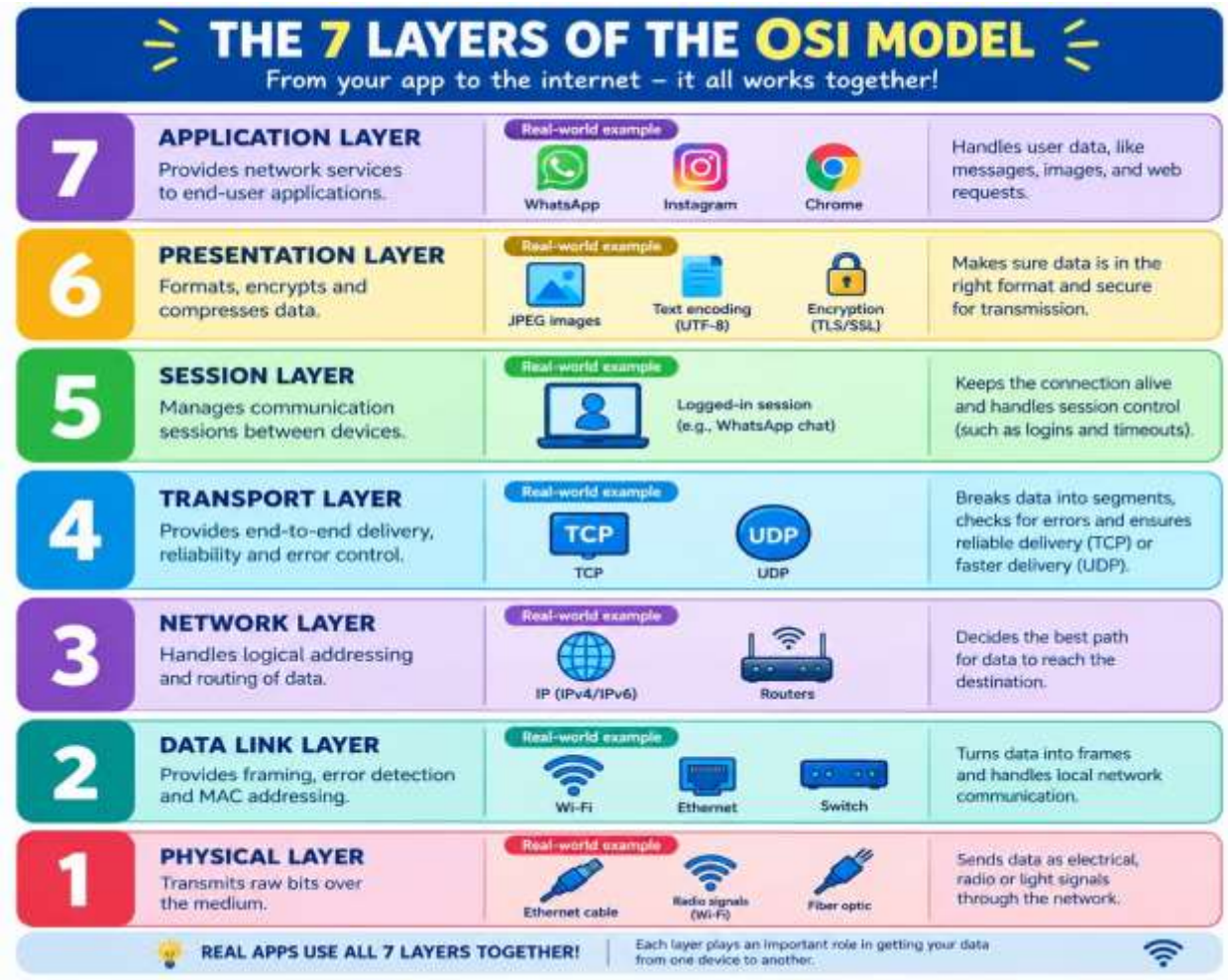
- OSI Layer: Layer 3 — Network
- Function: Provides IP addressing and routing.

STEP:

1. Opened the captured packets in Wireshark.
2. Inspected the Protocol column.
3. Selected three packets and examined their packet details.
4. Identified the main OSI layer associated with each protocol.

DAY-1 (NETWORKING)

TASK-3 Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



DAY-1 (NETWORKING)

TASK-4 Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.

Example: Sending the message "Hello" to a friend on WhatsApp.

1. Application Layer (Layer 7)

I type "Hello" in WhatsApp and press the Send button. WhatsApp prepares the message for communication.

2. Presentation Layer (Layer 6)

The message data is formatted and encrypted to help protect its contents from unauthorized access.

3. Session Layer (Layer 5)

The communication between devices is maintained so that data can be exchanged. Session-related functions may be handled by the application itself.

4. Transport Layer (Layer 4)

Transport protocols carry data between the devices. They help manage data delivery and communication between applications.

5. Network Layer (Layer 3)

The IP address and routing system help direct the data packets toward the receiving device through the internet.

6. Data Link Layer (Layer 2)

The data is placed into frames for transmission across the local network, such as Wi-Fi or Ethernet.

7. Physical Layer (Layer 1)

The data travels as radio signals, electrical signals, or light signals through the network medium.

At the Receiver's End:

The received data is processed through the network layers, and WhatsApp displays the message to my friend.